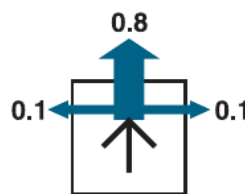


Roll: _____

Name: _____

Answer the questions (**Q1 to Q4**) based on the grid world. Consider the stochastic (4 x 3) Grid world where the “intended” outcome occurs with probability 0.8, but with probability 0.2 the agent moves at right angles to the intended direction. Transitions into any non-terminal state have a reward of -0.04. Assume that the rewards are undiscounted. Given the current estimate of utilities for states as follows:

3	0.8516	0.9078	0.9578	+1
2	0.8016		0.7003	-1
1	0.7453	0.6953	0.6514	0.4279
	1	2	3	4



(round numerical answers to 6 decimals)

[Q1] (NUM) (2 marks) Calculate the probability of the agent being in state (2, 1) after taking the *Up* action two times starting at state (1, 1).

[ANS]: _____

[Q2] (NUM) (4 marks) Calculate the utility for states (1, 1) and (3, 2).

[ANS]: $U(1, 1) =$ _____ $U(3, 2) =$ _____

[Q3] (NUM) (2 marks) Based on the new utility calculated for states (1, 1) and (3, 2) in the previous question, find out the *optimal actions* for those states i.e., the optimal actions to take when the agent is in that state. The action must be one of (Up, Down, Right, Left).

[ANS]: $\pi(1, 1) =$ _____ $\pi(3, 2) =$ _____

[Q4] (MCQ) (2 marks) We assumed the reward $R(s) = -0.04$. What should be the value of $R(s)$ such that an optimally acting agent will never reach a terminal state?

[A] $R(s) < 0$

[B] $R(s) = 0$

[C] $R(s) > 0$

[D] optimal agent will always reach a terminal state

[Q5] (MCQ) (2 marks) Assume that an MDP has an initial state S^* . According to the Markovian Property, the probability of reaching a state S' from a state S should depend:

- [A] on State S only
- [B] on State S' only
- [C] on State S^* only
- [D] on State S^* and S

[Q6] (MCQ) (2 marks) Fill in the blanks:

An optimal policy is the one which yields highest _____. In a/an _____ horizon MDP, the optimal policy is independent of the starting state if the rewards are _____.

- [A] reward, infinite, discounted
- [B] expected utility, finite, undiscounted
- [C] reward, finite, undiscounted
- [D] expected utility, infinite, discounted

[Q7] (MCQ) (3 marks) Select the correct bellman equation for Utility

[A] $U(s) = \max_a [R(s, a) + \gamma \sum_{s'} P(s'|a, s) U(s')]$

[B] $U(s) = \max_a \sum_{s'} P(s'|a, s) [R(s, a, s') + \gamma U(s')]$

[C] $U(s) = R(s) + \gamma \max_a \sum_{s'} P(s'|a, s) U(s')$

[D] All of the above are correct

[Q8] {MSQ} (4 marks) Use the provided truth table to select correct logical entailments.

p	q	r	$p \rightarrow q \vee r$	$p \rightarrow r$	$q \rightarrow r$
1	1	1	1	1	1
1	1	0	1	0	0
1	0	1	1	1	1
1	0	0	0	0	1
0	1	1	1	1	1
0	1	0	1	1	0
0	0	1	1	1	1
0	0	0	1	1	1

- [A] $\{p \rightarrow q \vee r\} \models (p \rightarrow r)$
- [B] $\{p \rightarrow r\} \models (p \rightarrow q \vee r)$
- [C] $\{q \rightarrow r\} \models (p \rightarrow q \vee r)$
- [D] $\{p \rightarrow q \vee r, q \rightarrow r\} \models (p \rightarrow r)$

[Q9] (MCQ) (2 marks) An inference algorithm that derives only entailed sentences is called _____ and if it can derive any sentence that is entailed, it is called _____. Hence, if KB is true in the real world, then any sentence α derived from KB by a _____ inference procedure is also true in the real world.

- [A] sound, complete, sound
- [B] complete, truth-preserving, sound
- [C] truth-preserving, sound, complete
- [D] complete, sound, truth-preserving

[Q10] {MSQ} (2 marks) Let Γ and Δ be sets of sentences in Propositional Logic, and let ϕ and ψ be individual sentences in Propositional Logic. Select the correct statements from the following:

- [A] If $\Gamma \models \phi$ and $\Delta \models \phi$, then $\Gamma \cap \Delta \models \phi$
- [B] If $\Gamma \models \phi$ and $\Delta \models \phi$, then $\Gamma \cup \Delta \models \phi$
- [C] If $\Gamma \models \phi$ and $\Delta \not\models \phi$, then $\Gamma \cup \Delta \models \phi$
- [D] If $\Gamma \not\models \psi$, then $\Gamma \models \neg \psi$